Lesson 04 Demo 02

Demonstrating Nested Routing in a React Application

Objective: To demonstrate the implementation of nested routes by creating a structured and hierarchical navigation system in a React application using Vite and React Router

Tools required: Visual Studio Code and Node.js

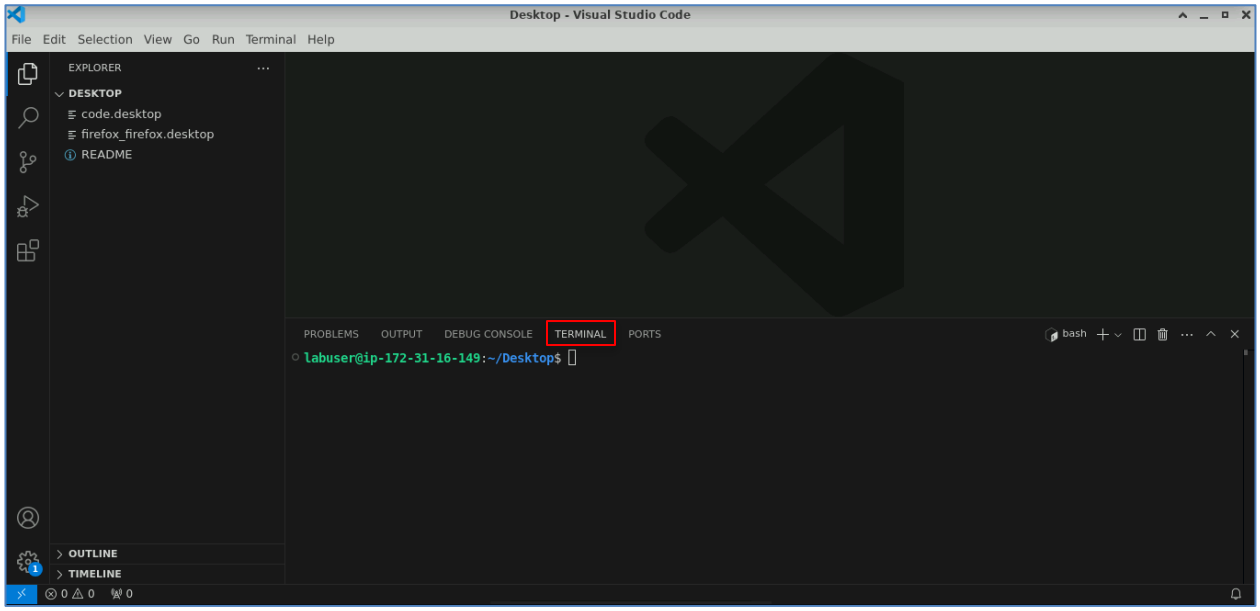
Prerequisites: None

Steps to be followed:

1. Set up a new React project using Vite
2. Create React components
3. Import and use the components
4. Run and verify the application

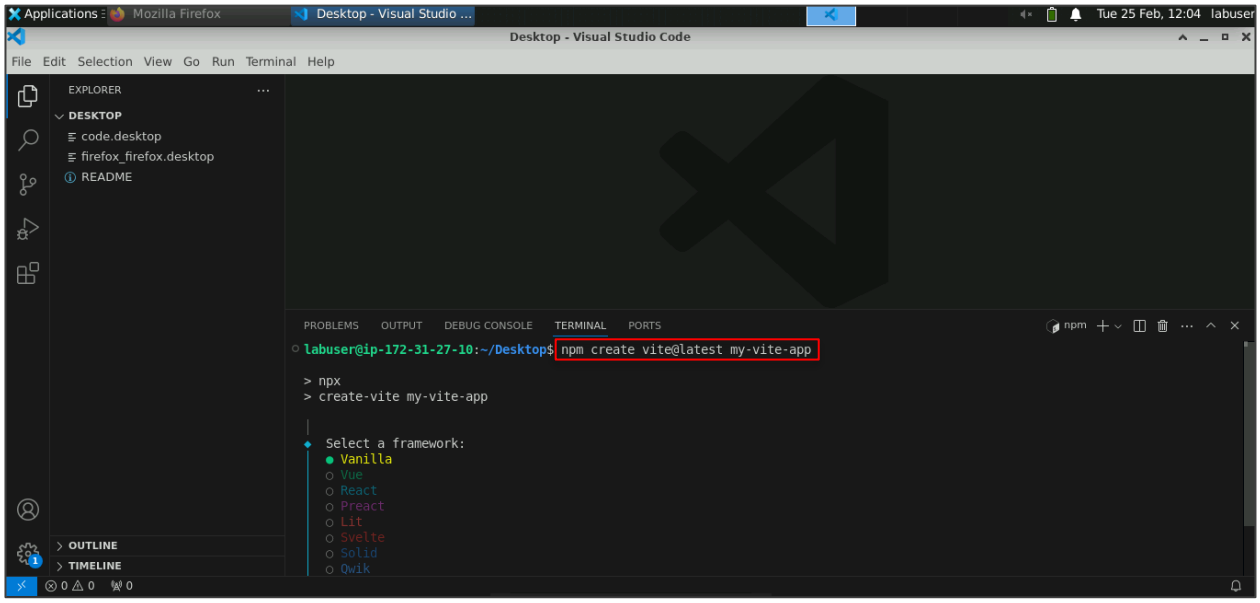
Step 1: Set up a new React project using Vite

1.1 Navigate to VS Code and open the terminal

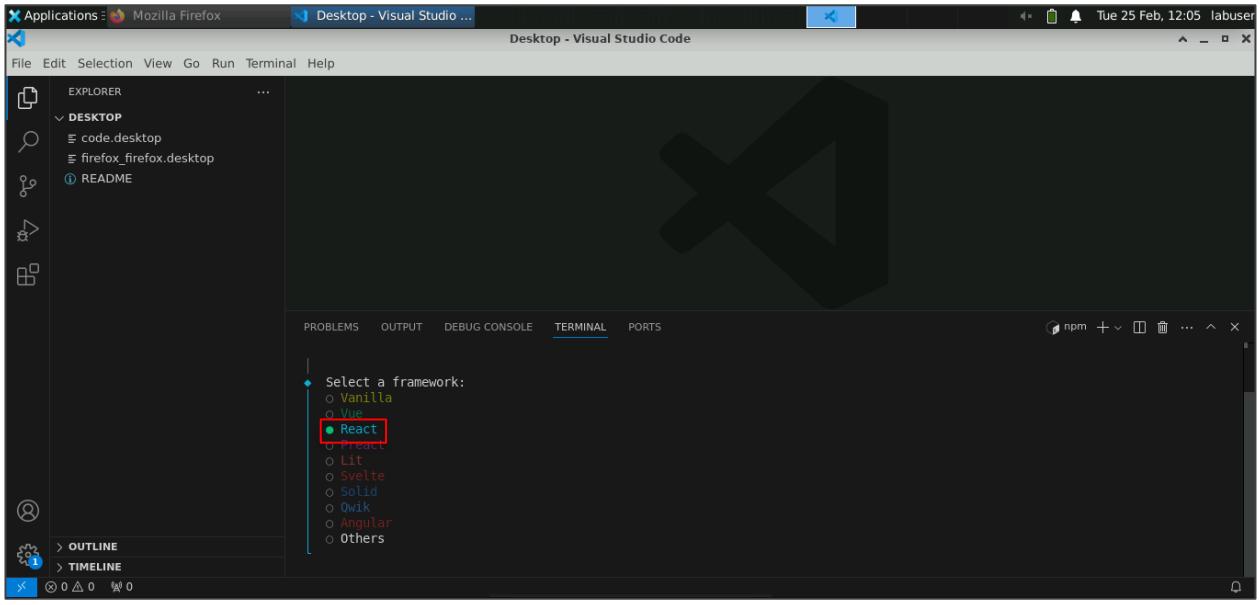


1.2 Run the following command to create a new React project using Vite:

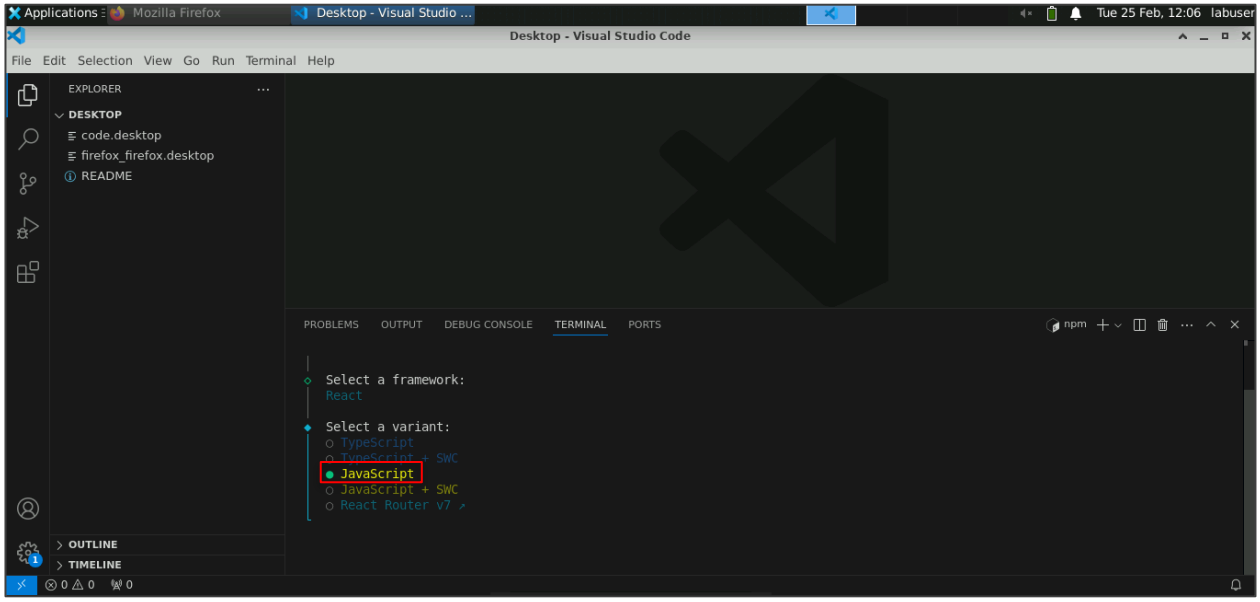
npm create vite@latest my-vite-app



1.3 Select **React** and press enter

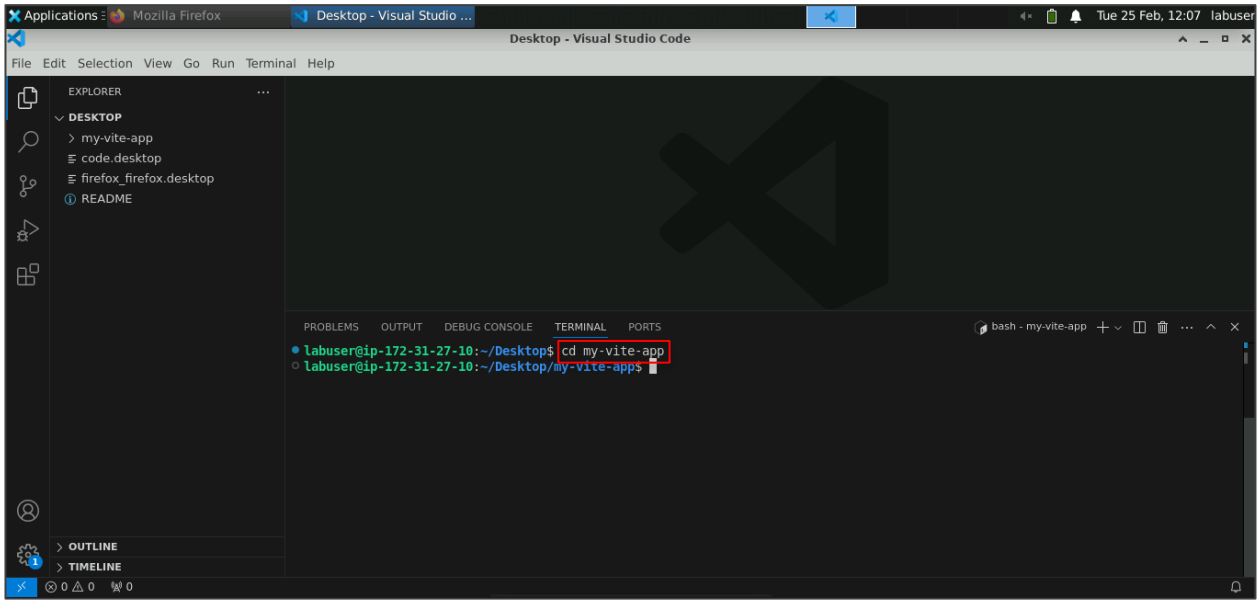


1.4 Select **JavaScript** and press enter



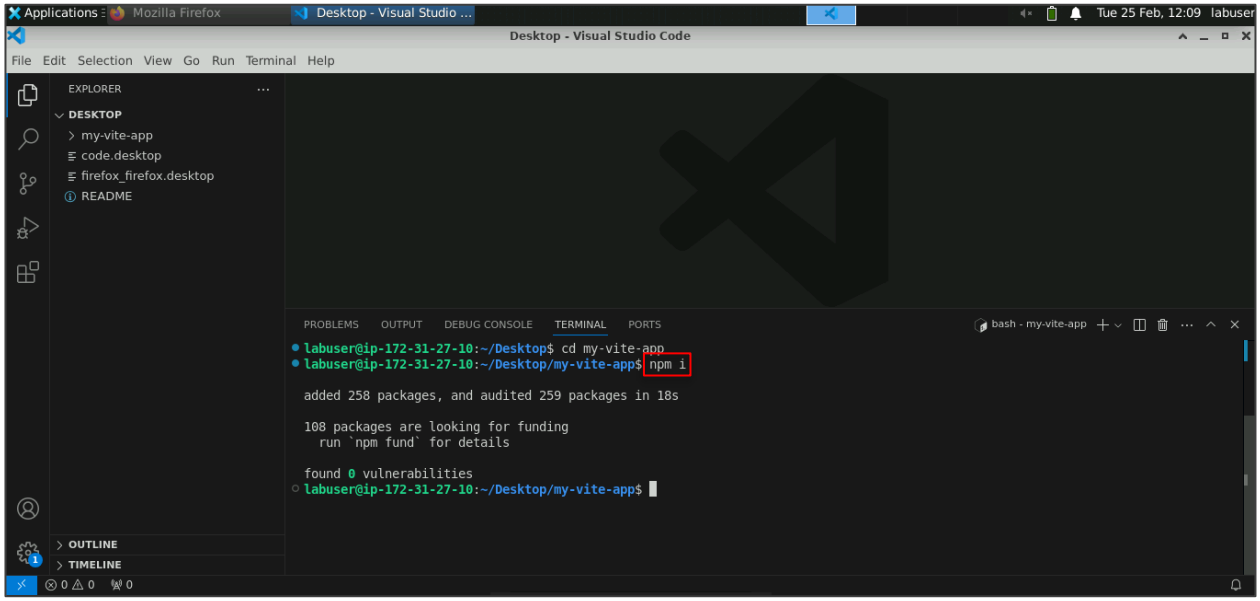
1.5 Run the following command to navigate to the project folder:

cd my-vite-app



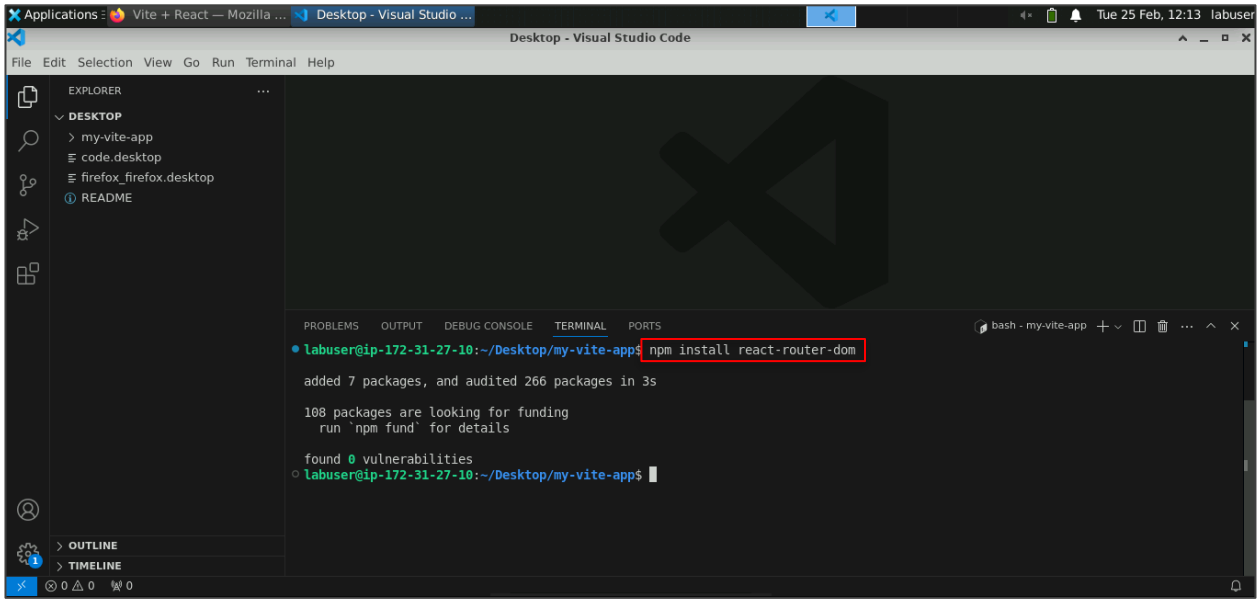
1.6 Run the following command to install dependencies:

npm i



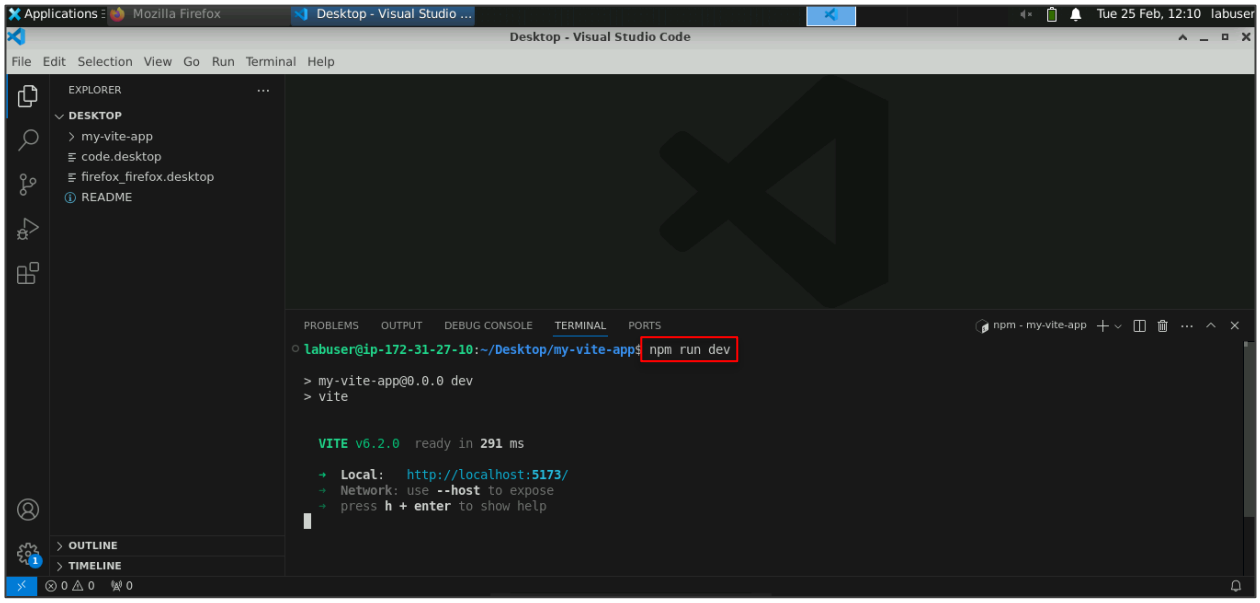
1.7 Run the following command to install the React router:

npm install react-router-dom



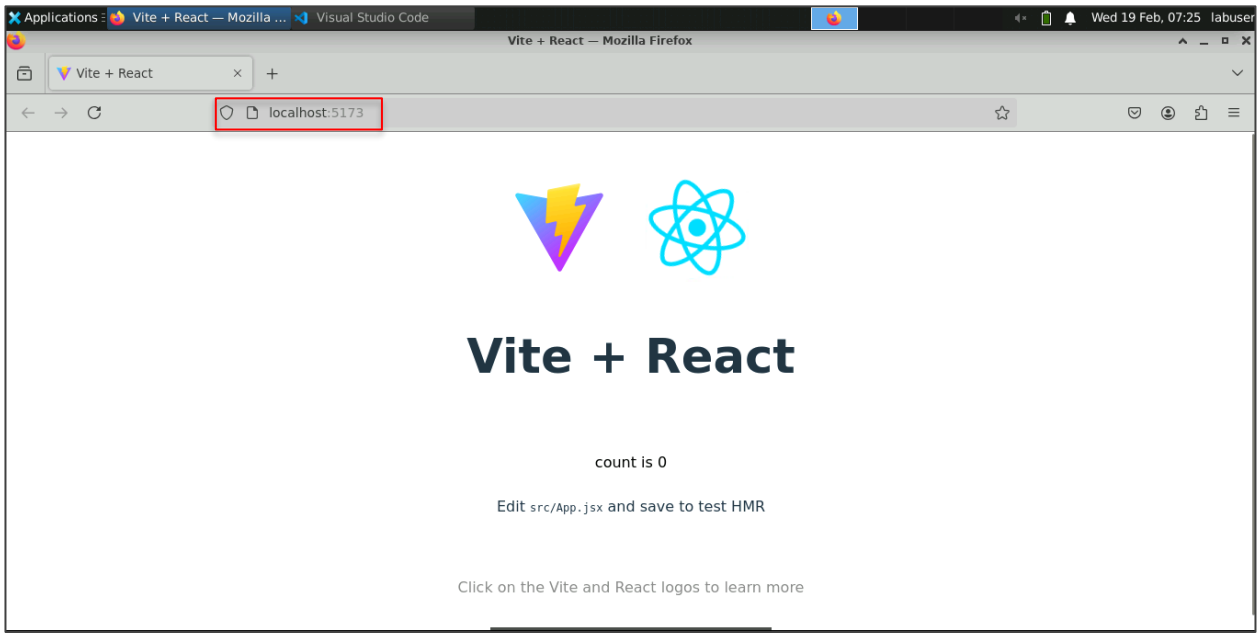
1.8 Run the following command to start the development server:

npm run dev



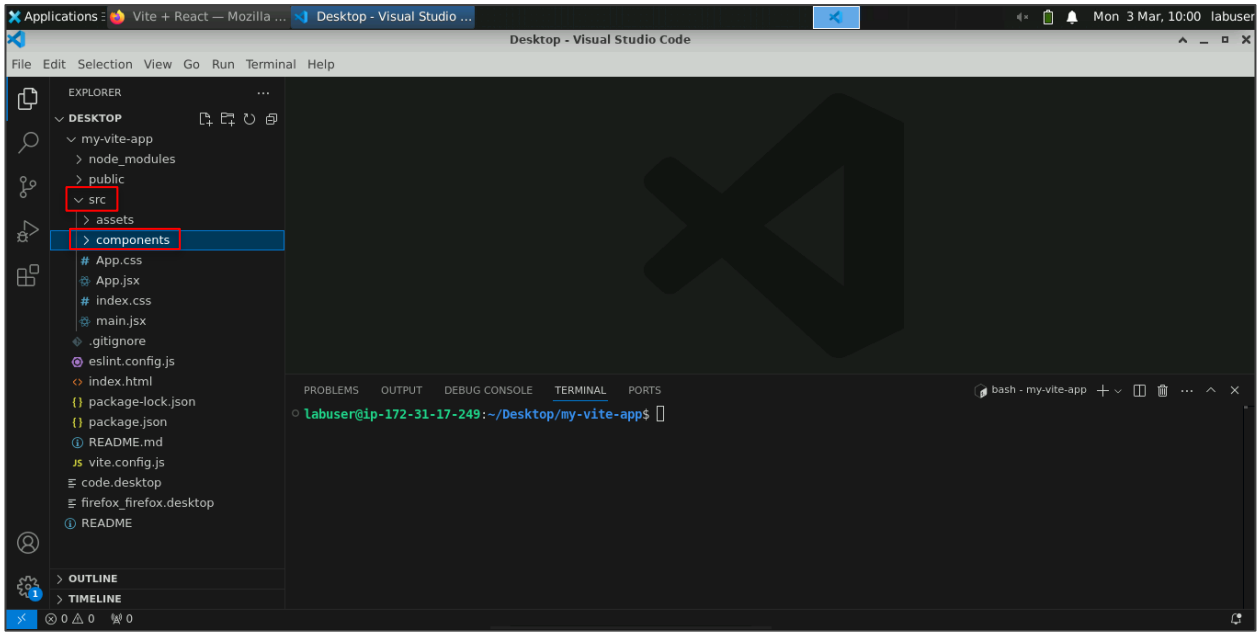
1.9 Navigate to the web browser and paste the link below:

http://localhost:5173

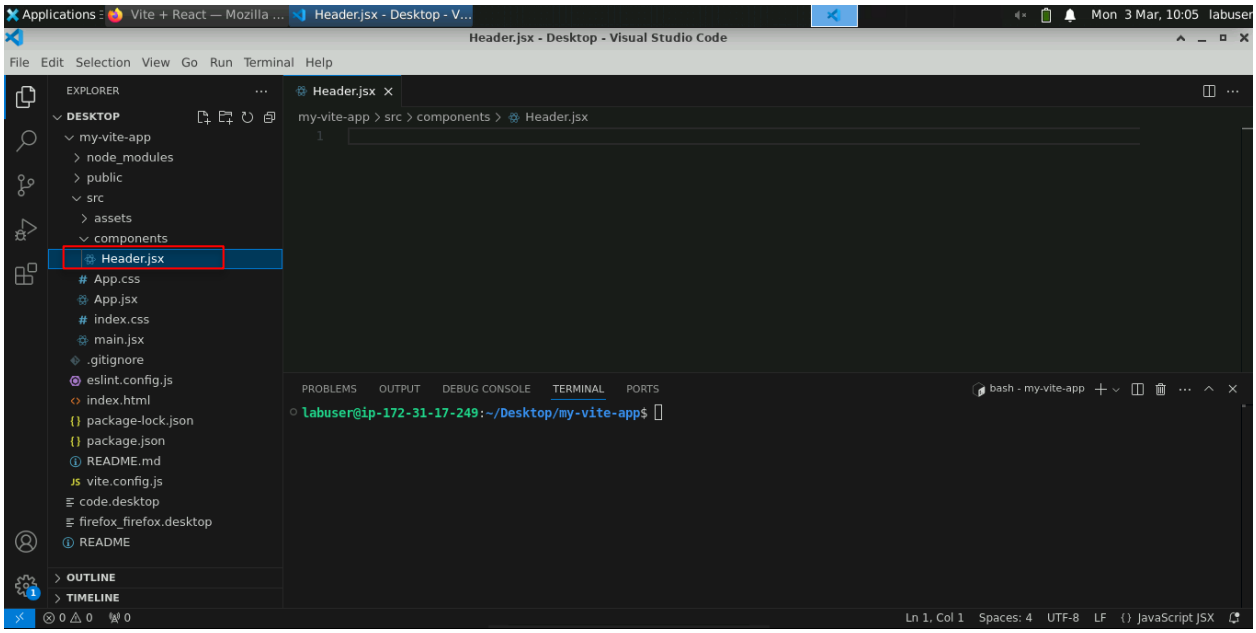


Step 2: Create React components

2.1 Inside the **src** folder, create a new folder named **components**

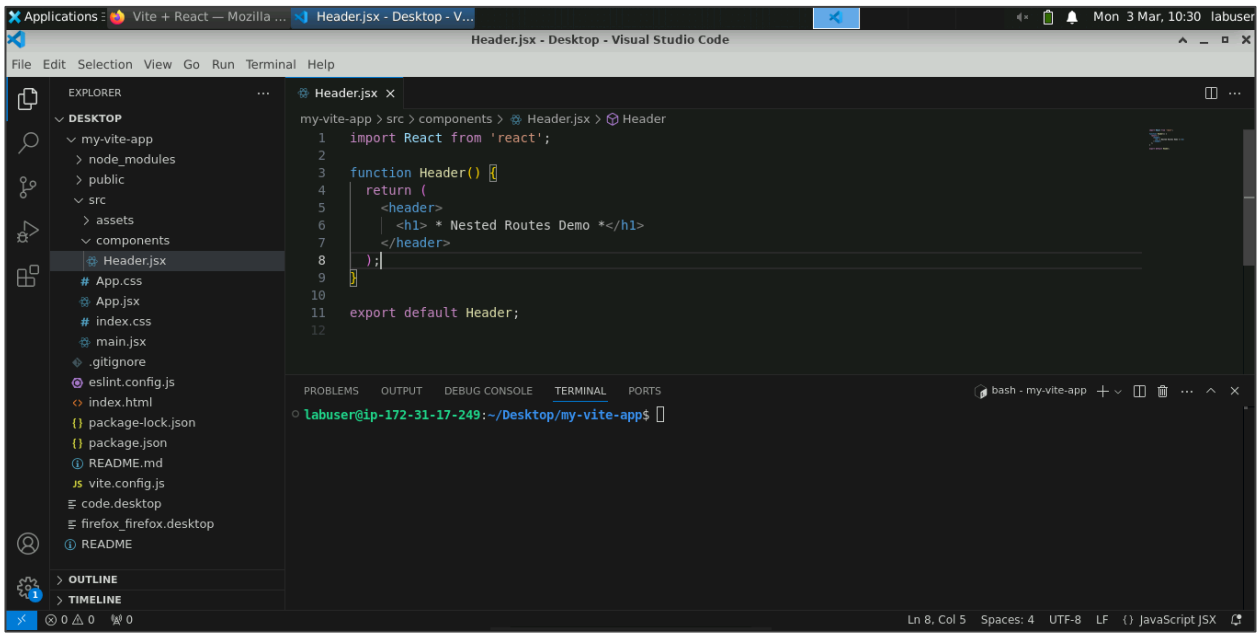


2.2 Inside the **components** folder, create a new file named **Header.jsx**

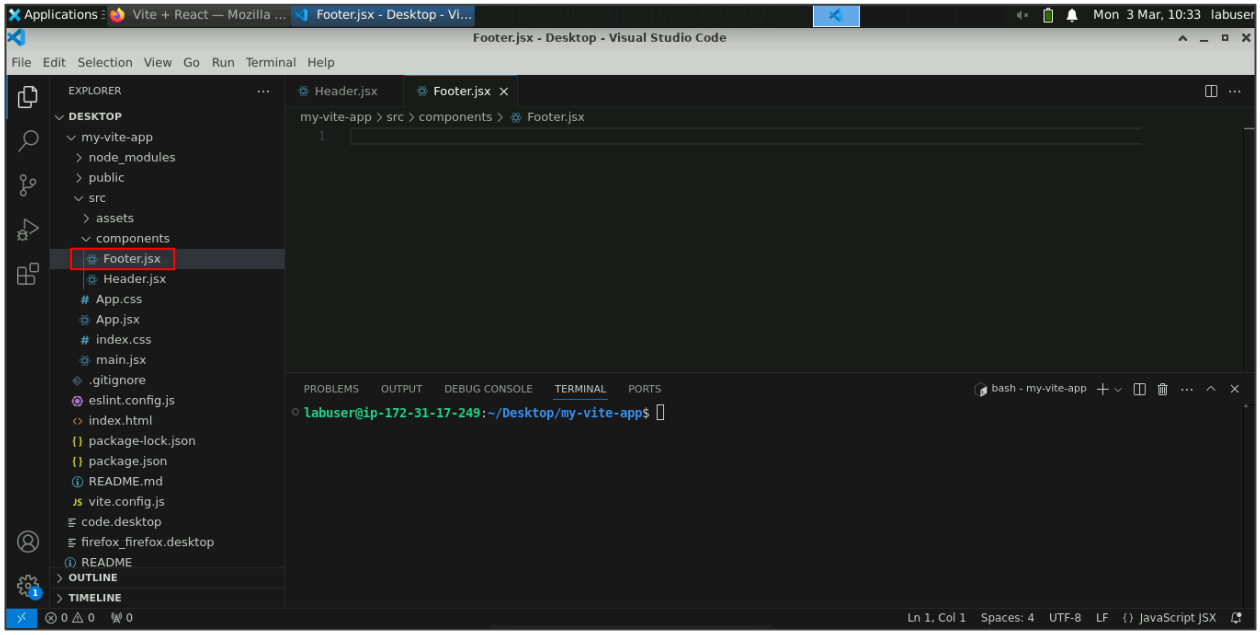


2.3 Add the following code to **Header.jsx:**

```
import React from 'react';

function Header() {
  return (
    <header>
      <h1> * Nested Routes Demo *</h1>
    </header>
  );
}

export default Header;
```

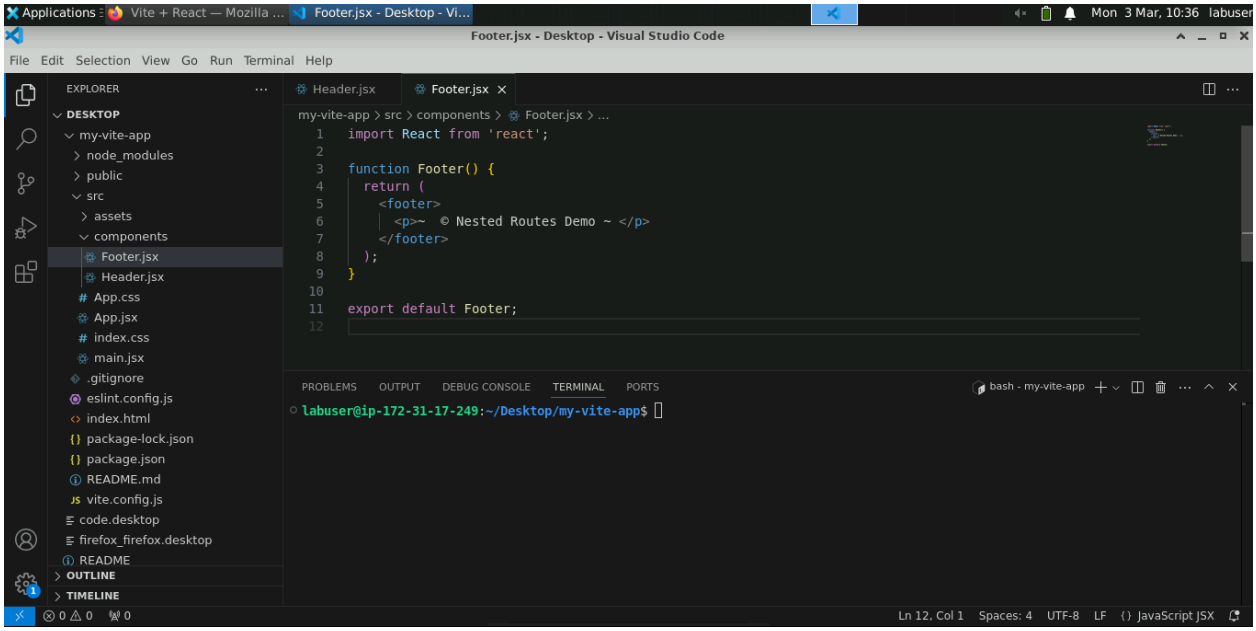


2.4 Inside the **components** folder, create a new file named **Footer.jsx**

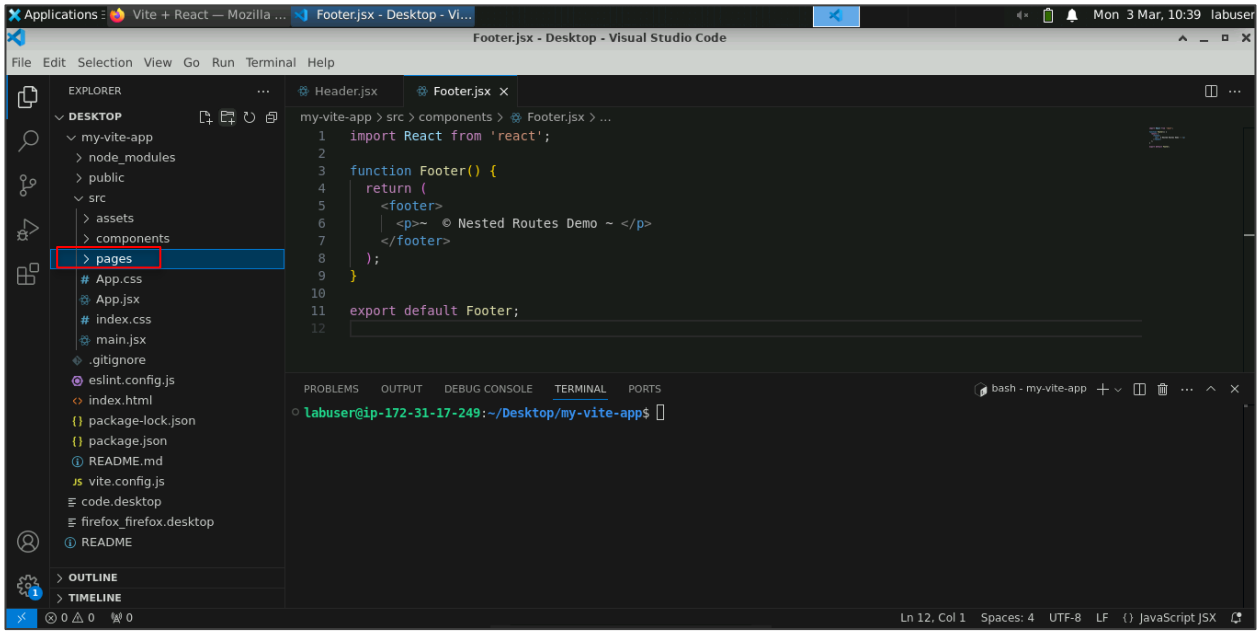


2.5 Add the following code to **Footer.jsx:**

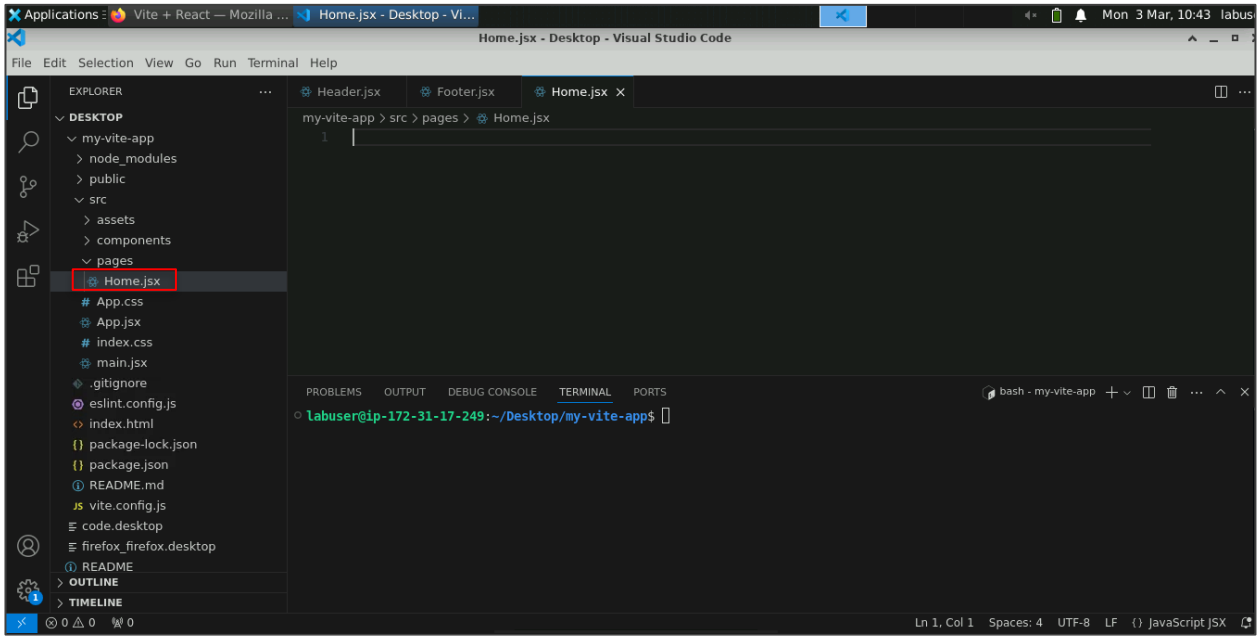
```
import React from 'react';

function Footer() {
  return (
    <footer>
      <p>~  © Nested Routes Demo ~ </p>
    </footer>
  );
}

export default Footer;
```



2.6 Inside the **src** folder, create a new folder named **pages**

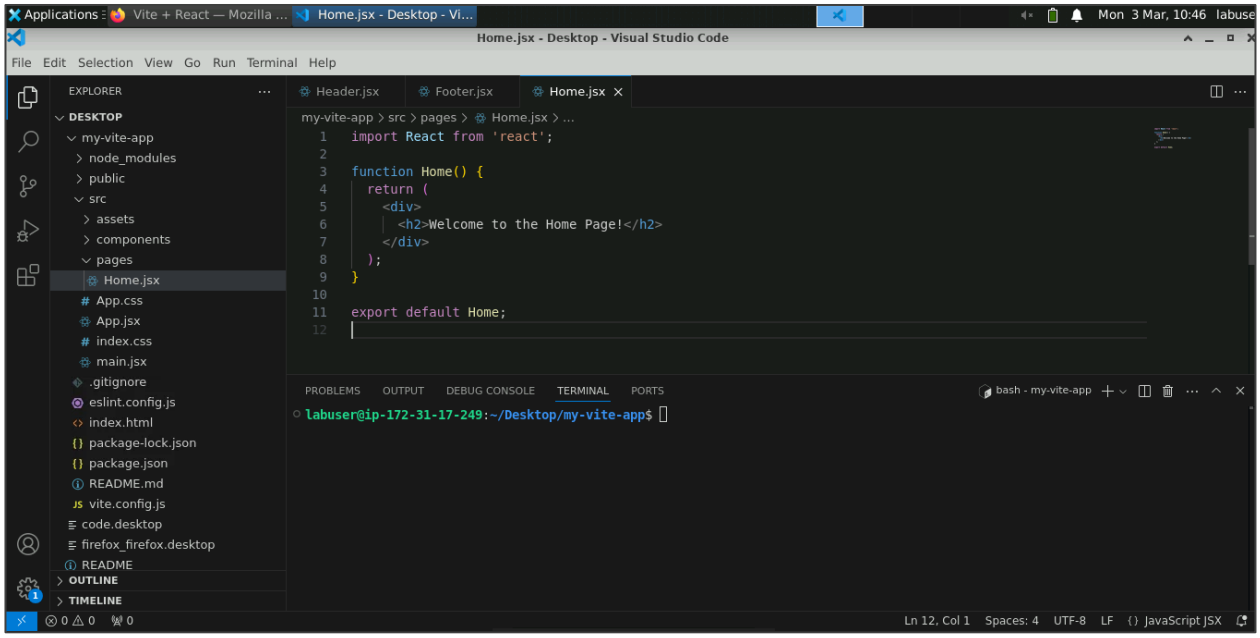


2.7 Inside the **pages** folder, create a new file named **Home.jsx**

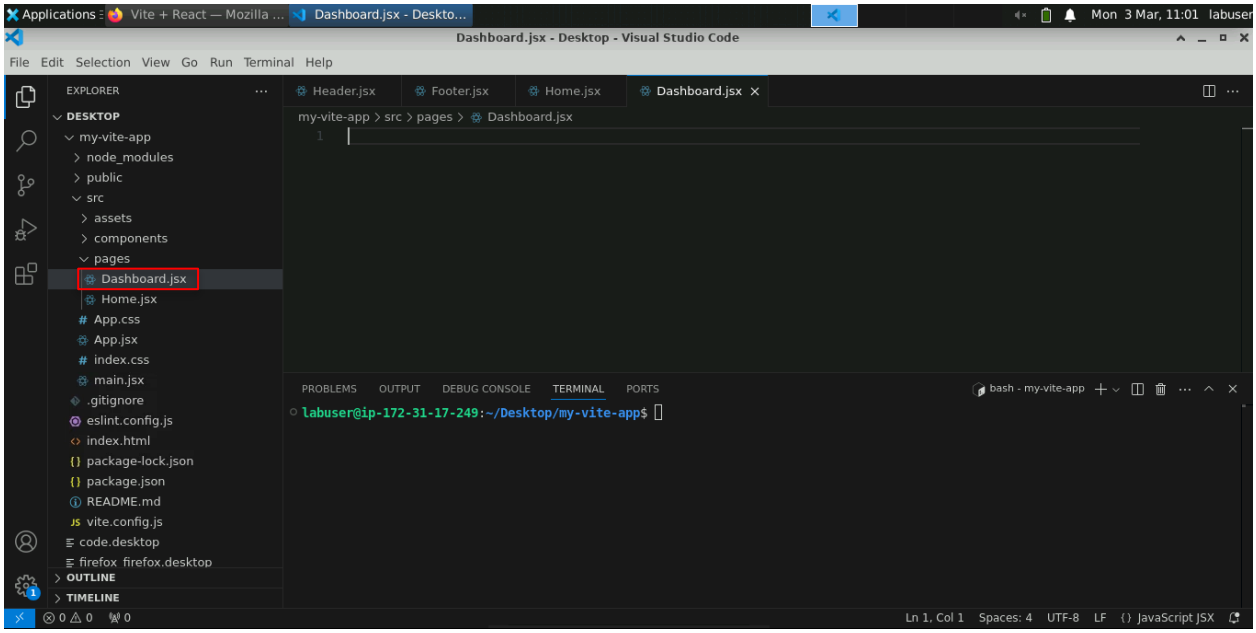


2.8 Add the following code to **Home.jsx:**

```
import React from 'react';

function Home() {
  return (
    <div>
      <h2>Welcome to the Home Page!</h2>
    </div>
  );
}

export default Home;
```

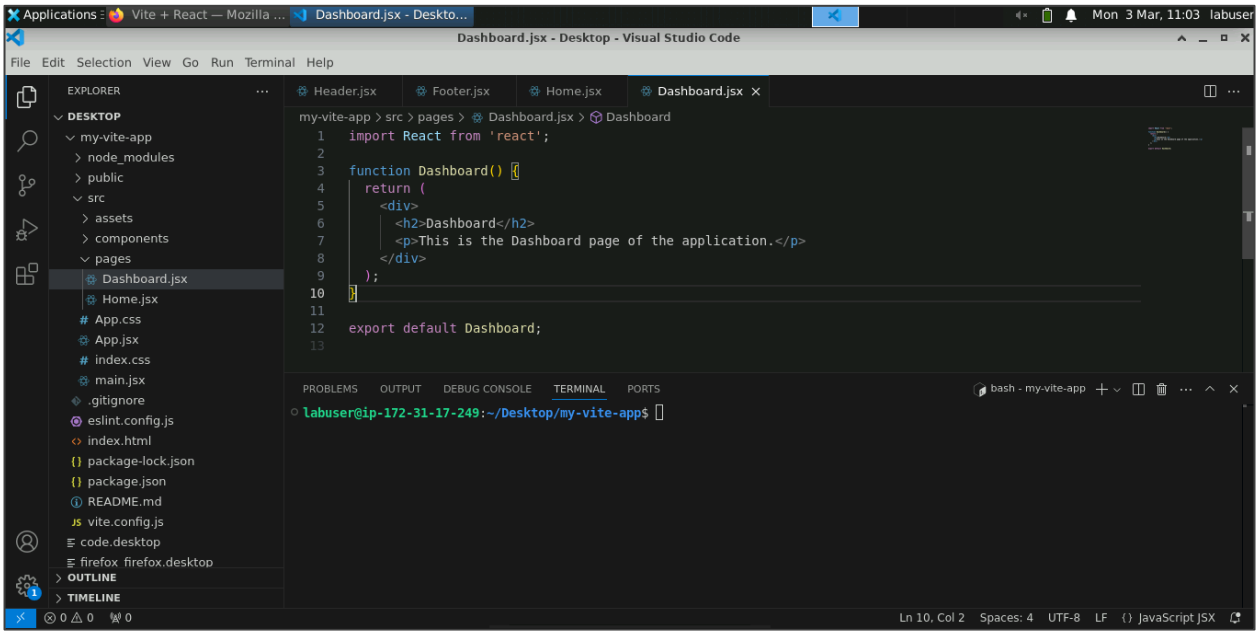


2.9 Inside the **pages** folder, create a new file named **Dashboard.jsx**



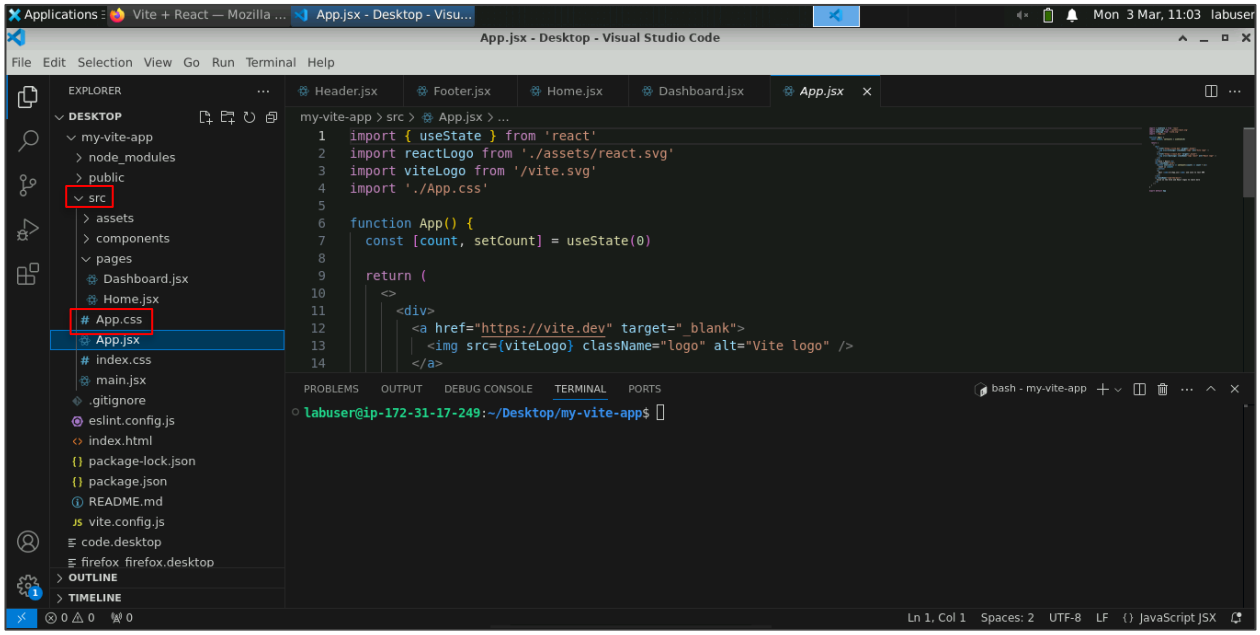
2.10 Add the following code to **Dashboard.jsx:**

```
import React from 'react';

function Dashboard() {
  return (
    <div>
      <h2>Dashboard</h2>
      <p>This is the Dashboard page of the application.</p>
    </div>
  );
}

export default Dashboard;
```



Step 3: Import and use the components

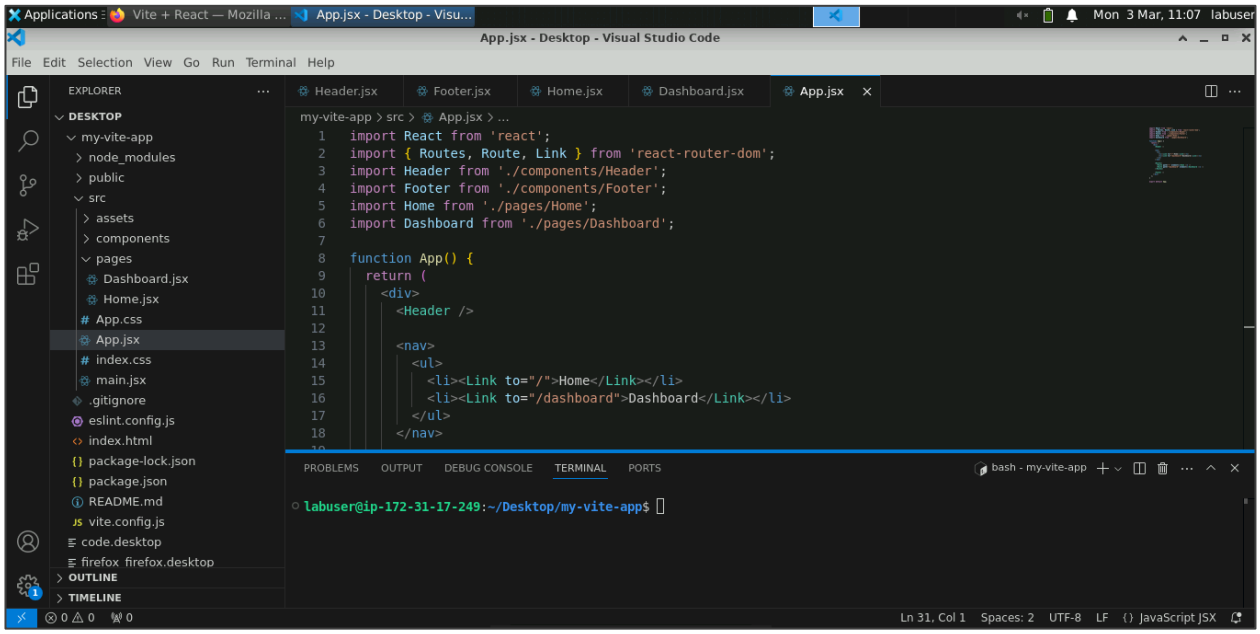
3.1 Open the **App.jsx** file inside the **src** folder



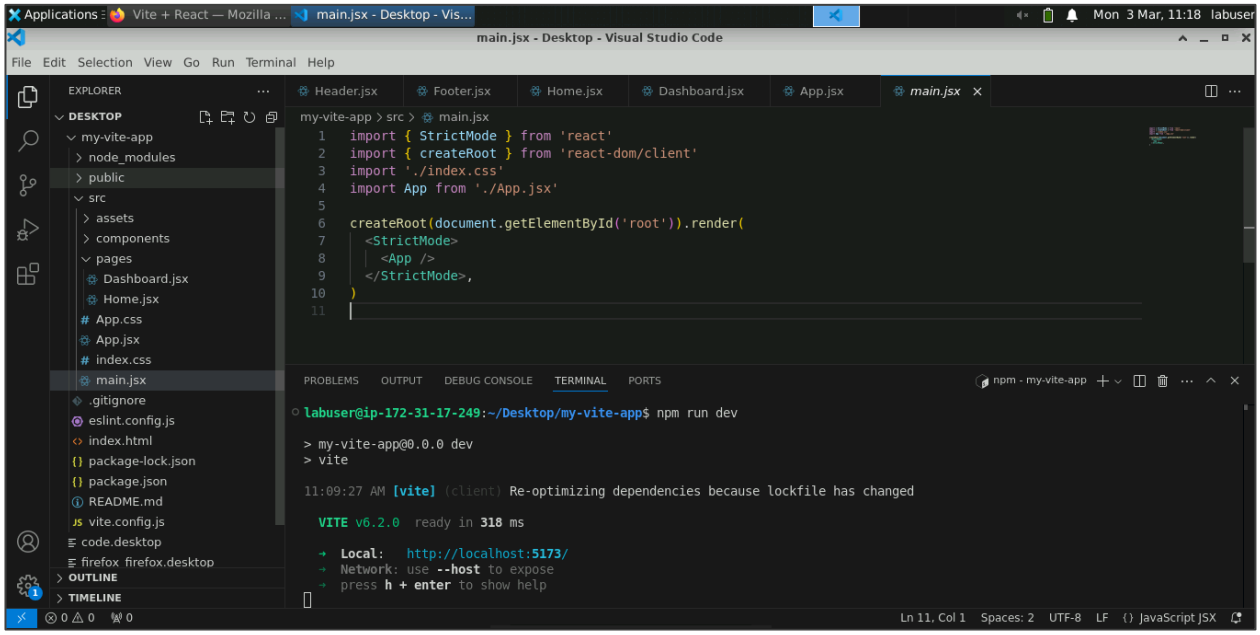
3.2 Paste the code below to include the new components in **App.jsx**:

```
import React from 'react';
import { Routes, Route, Link } from 'react-router-dom';
import Header from './components/Header';
import Footer from './components/Footer';
import Home from './pages/Home';
import Dashboard from './pages/Dashboard';

function App() {
  return (
    <div>
      <Header />

      <nav>
        <ul>
          <li><Link to="/">Home</Link></li>
          <li><Link to="/dashboard">Dashboard</Link></li>
```

```
      </ul>
    </nav>

    <Routes>
      <Route path="/" element={<Home />} />
      <Route path="/dashboard" element={<Dashboard />} />
    </Routes>

    <Footer />
  </div>
 );
}

export default App;
```

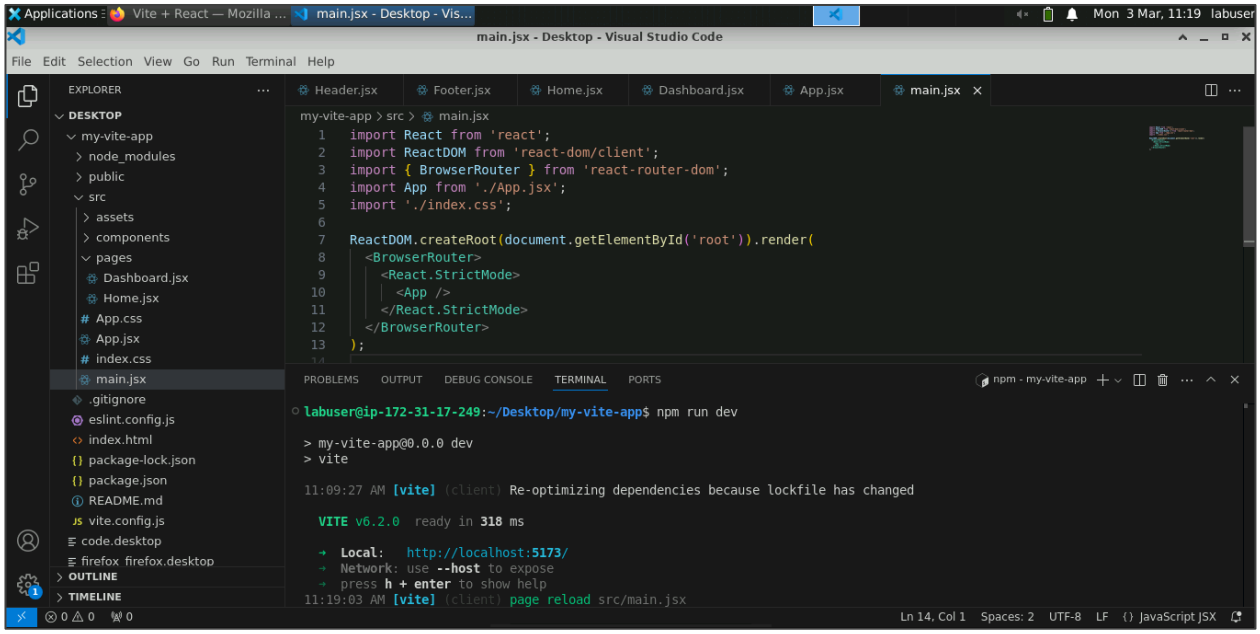


3.3 Open the **main.jsx** file inside the **src** folder



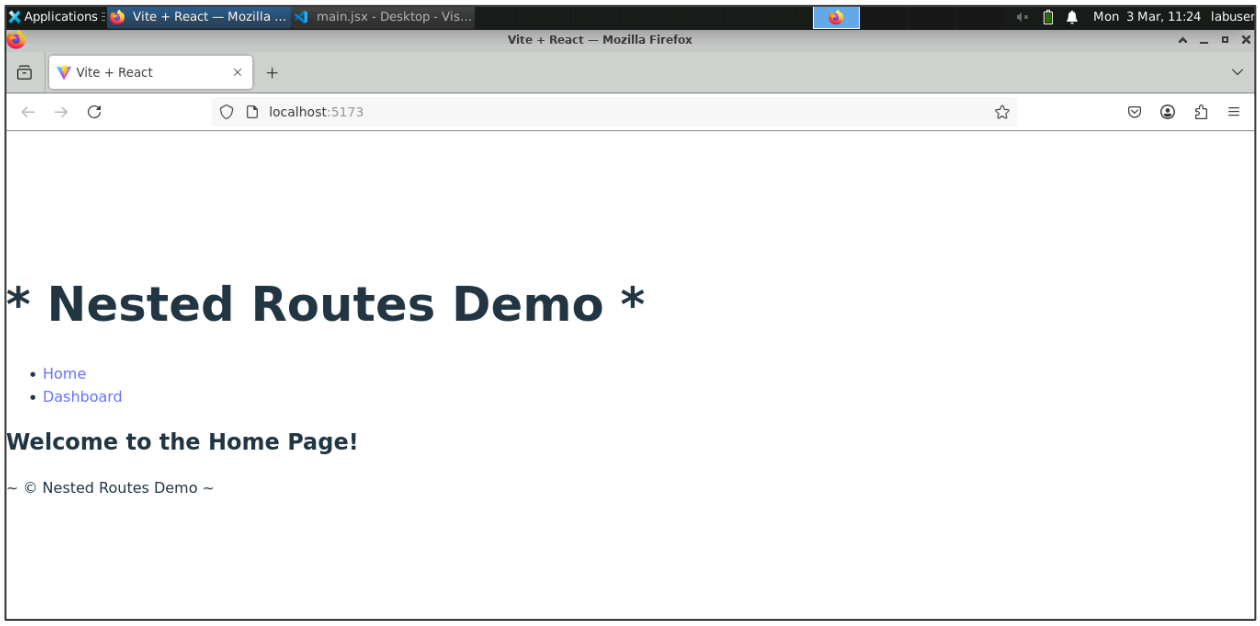
3.4 Paste the code below to include the new components in **main.jsx**:

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import { BrowserRouter } from 'react-router-dom';
import App from './App.jsx';
import './index.css';

ReactDOM.createRoot(document.getElementById('root')).render(
  <BrowserRouter>
    <React.StrictMode>
      <App />
    </React.StrictMode>
  </BrowserRouter>
);
```

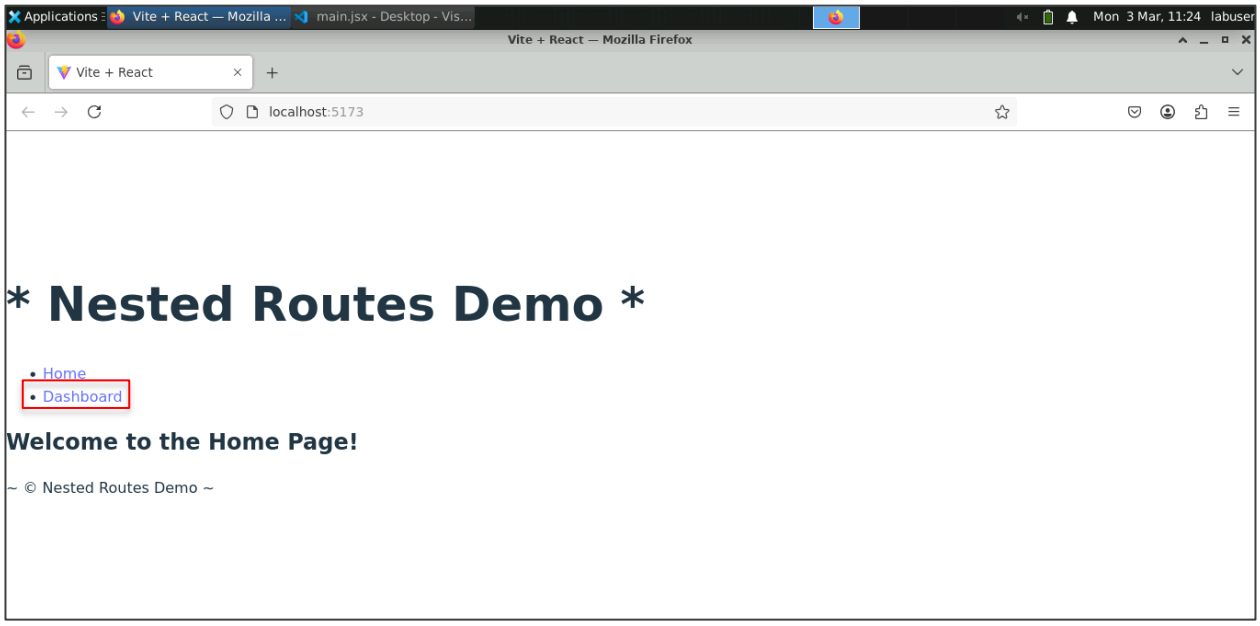


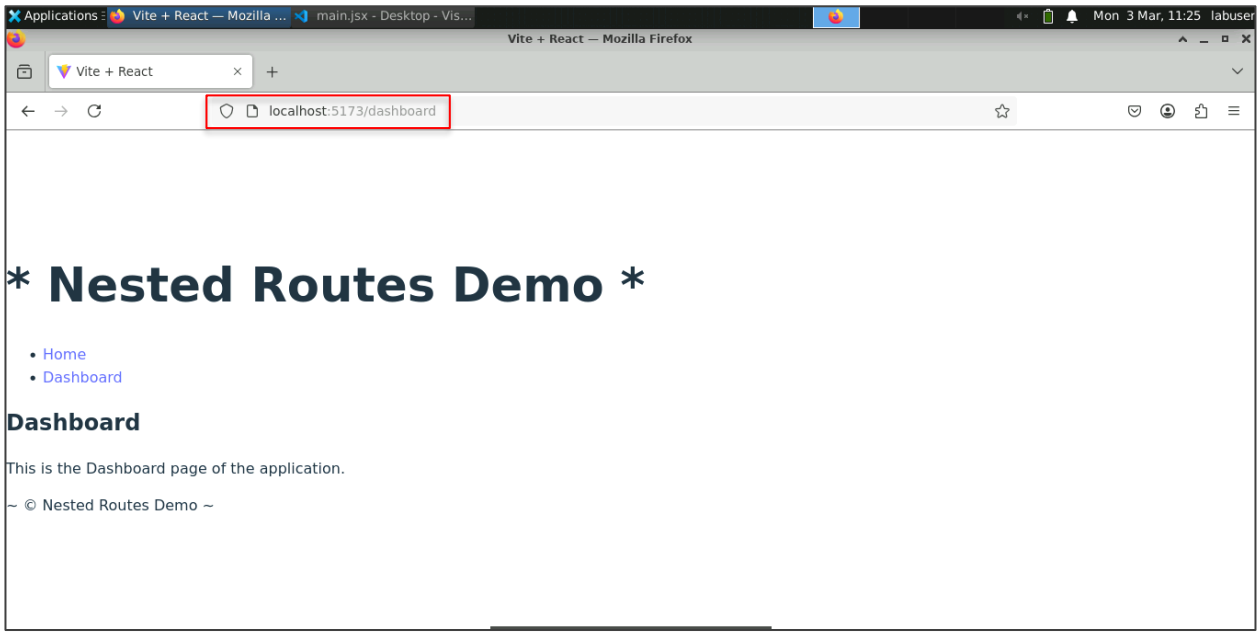
Step 4: Run and verify the application

4.1 Navigate to the browser and refresh the webpage



4.2 Click **Dashboard** to navigate to **/dashboard** route





By following these steps, you have successfully implemented nested routing in a React application using React Router with Vite, enabling smooth navigation and rendering between different pages or components.